# LING406_Lecture12_Notes

Course: LING406: Psycholinguistics

Title of Lecture: Lecture 11: Speech Perception I: Categorical Perception

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**Section 1: A Very Old and Very General Model of Speech Perception**

- **Categorical perception** is the tendency to hear a continuously varying speech signal as a small number of discrete sound categories. The analog signal comes in through the **cochlea**, and by the time it activates **phonemes**, a lot of information has been **lost**. The listener no longer keeps track of exactly how a sound was produced. The listener keeps only which category the sound belongs to.
- This **loss of information is helpful**. When the system throws away acoustic detail that does not separate one category from another, it can make contact with **higher levels of representation**, such as words. A natural assumption follows from this: **in order to identify words, we first need to identify their constituent phonemes**.
- **Klatt (1979)**, published in the *Journal of Phonetics*, described speech perception as a fixed series of stages. Each stage takes in one kind of representation and passes a transformed version to the next stage.
    1. The **speech waveform** enters the **peripheral auditory system**, which performs a **spectral analysis** and breaks the sound into the frequencies it contains.
    2. The spectral analysis feeds a set of **acoustic property detectors**. Their output is **detector outputs versus time**, a record of which acoustic properties are present at each moment.
    3. Those detector outputs feed **phonetic feature detectors**. Their output is **feature probabilities versus time**, an estimate of how likely each phonetic feature, such as voicing, is at each moment.
    4. The feature probabilities feed **segmental analysis**, which produces a **segmental feature matrix**. This is a sequence of segments, or phonemes, each described by its set of features.
    5. The segmental feature matrix feeds **lexical search**, which produces **lexical hypotheses**, the candidate words that match the sequence of segments.
- In this model, each stage hands a finished product to the next stage, and word recognition starts only after the phonemes have been identified. Two problems challenge this simple view: **cascaded processing** and **feedback**.
    - **Cascaded processing** challenges the idea that each stage has to finish its work before the next stage can start.
    - **Feedback** challenges the idea that information only flows upward, from sound to phonemes to words, and never back down.

**Section 2: A Voicing Continuum from "Peach" to "Beach"**

- A **voicing continuum** is a series of synthetic speech sounds that changes in small, equal steps from one word to another along a single acoustic dimension. This continuum has nine steps running between "peach" and "beach." The dimension that changes is **voice onset time (VOT)**, which is the delay between the release of a stop consonant and the moment the vocal folds start vibrating. A long VOT is heard as the voiceless /p/, and a short VOT is heard as the voiced /b/. Making the VOT **shorter** moves the sound from "peach" toward "beach."

- Each step can be displayed as a **spectrogram**, with frequency from 0 to 5000 Hz on the vertical axis, time from 0 to 0.5 seconds on the horizontal axis, and darkness representing how much energy is present.
    - The dark horizontal bands are **formants**, the resonant frequencies of the vocal tract, and they mark the vowel.
    - The fine vertical stripes inside those bands are individual pulses of the vocal folds, so they mark **voicing**.
    - After the vowel there is a short blank gap of silence followed by a patch of scattered noise, and that noise is the "ch" sound at the end of both words.
    - The spectrograms for "peach," for the middle steps, and for "beach" look nearly identical. The only thing that changes across the continuum is the timing of voicing at the very start of the word.
- In the **identification task**, listeners hear one step at a time and label it as one word or the other. The identification proportion stays at 1 for steps 1 through 5, falls to about 0.33 at step 6, and drops to 0 for steps 7 through 9. Steps 1 through 5 receive the same label on every trial, and steps 7 through 9 receive the other label on every trial.
    - Step 6 is circled in red because it is the only ambiguous step. It sits right at the **category boundary**, and listeners give it one label on about a third of trials and the other label on the rest.
- In the **discrimination task**, listeners hear two steps that are two positions apart and judge whether they are the same or different. The pairs 1–3, 2–4, and 3–5 are discriminated at 0. The pairs 4–6 and 5–7 are discriminated at 1. The pair 6–8 is discriminated at about 0.33, and the pair 7–9 drops back to 0.
    - The physical difference is exactly the same for every pair, since every pair is two steps apart.
    - Pairs inside one category cannot be told apart at all, while pairs that straddle the boundary are told apart perfectly.
    - The 6–8 pair is discriminated at about 0.33, which matches the roughly one-third of trials on which step 6 is labeled differently from step 8.
- This pattern is the core of **categorical perception**. How well two sounds can be distinguished depends on whether they get different category labels, not on how different they are acoustically.

**Section 3: How Categorical Perception Could Happen: The Threshold Model**

- In a **threshold model**, activity is only passed forward from one level to the next when an **activity threshold** is crossed. A lower level can be partly active for a long time, but the level above it receives nothing until that activity reaches a fixed cutoff. In this kind of model, categorical perception arises at the stage of **phoneme identification**.
- At the feature level, the activation of the voicing feature rises and falls like a wave across 100 time steps. Two dashed lines mark the thresholds, one at about +1.25 for **+voiced** and one at about −1.25 for **−voiced**.
    - The wave first peaks at about 1.05 and then at about 1.2, so both peaks stay below the +voiced threshold.
    - The wave dips to about −0.85 and then about −0.7, so both dips stay above the −voiced threshold.
    - Only at the very end does the wave climb to about 1.4 and cross the +voiced threshold.
- At the phoneme level, activation stays completely flat at 0 for almost the entire time, even though the feature level is swinging up and down. Right around time step 97, when the feature level finally crosses the +voiced threshold, activation for /**b**/ jumps suddenly to about 1.5. The /p/ threshold at the bottom is never reached.

    - The phoneme level receives no information at all while the feature level is undecided.
    - Once the threshold is crossed, the phoneme is either fully on or fully off, so the output is **all-or-none**.
    - Graded differences at the feature level never reach the phoneme level, which is why the phoneme level behaves categorically.
- In this model, two tokens that are both eventually labeled /b/ look identical to every level above the phoneme, even if one of them had much weaker evidence for voicing than the other.

**Section 4: How Categorical Perception Could Happen: The Cascaded Model**

- In a **cascaded model**, activity is passed forward **continuously over time** from one level to the next. A lower level does not have to reach a cutoff before the level above it starts responding. In this kind of model, categorical perception might happen later on in processing, or it might even be an **artifact of the task structure**.
- With the same feature-level input as before, phoneme activation now follows the feature activation moment by moment. It rises to about 1.0, dips to about −0.9, rises to about 1.1, dips to about −0.85, and ends at about 1.2, just under the /b/ threshold. The phoneme-level curve is a slightly weaker copy of the feature-level curve.
    - Partial evidence for voicing produces partial activation of /b/.
    - Partial evidence against voicing produces partial activation in the direction of /p/.
    - The levels above the phoneme therefore receive graded information, including how good an example of /b/ a particular sound is.
- The idea that categorical perception could be an **artifact of the task structure** comes from the way identification tasks are set up. The listener is forced to choose one of two labels on every trial. Even if the internal representation is graded, the required answer is discrete, so the response curve comes out step-shaped. A step-shaped identification function could reflect the decision the task demands rather than the form of the representation underneath.

**Section 5: A Metaphor for Thresholded Activity: Making Furniture**

- Thresholded processing works like a chain of production in which each step takes a finished product from the step before.
    1. A **lumber mill** takes **trees** as input and produces **lumber** as output.
    2. A **furniture factory** takes **lumber** as input and produces **unassembled furniture** as output, meaning flat wooden pieces, legs, round tabletops, screws, and hardware.
    3. An **IKEA customer** takes **unassembled furniture** as input and produces finished **furniture**, such as a round table, as output.
- Each "layer" of this system can only take a specific input that has been transformed **completely** by the previous layer. The factory has to wait for the shipment of lumber before it can start manufacturing. The customer cannot start assembling until the boxed parts arrive, and a half-cut tree is of no use to the factory.
- In thresholded speech perception, the phoneme level waits for the feature level to deliver a finished decision, and the word level waits for finished phonemes. Nothing partial is ever passed along.
- **Many computer programs function this way**. One routine runs to completion and then hands its output to the next routine.

**Section 6: A Metaphor for Cascaded Processing: A Food Web**

- Cascaded processing works more like a **desert food web**, where arrows run from each food source to the animals that eat it.
    - **Desert plants** are eaten by **insects**, **small lizards**, and **kangaroo rats**.
    - Insects are eaten by **scorpions** and **large lizards**, and small lizards are also eaten by large lizards.
    - Large lizards, **snakes**, and kangaroo rats in turn feed the top predators, **hawks** and **desert foxes**.
- The "levels" of a food web **continuously interact** with one another. A drought that reduces the number of plants immediately affects the health of the kangaroo rats, which then affects the snakes, and so on up the web. No level waits for another level to finish anything, and change spreads across the system continuously.
- A food web also contains clear **feedback loops**. Adding more hawks changes the populations of snakes, lizards, and rats, so influence also travels from the top of the web back down.
- The central question is whether **phoneme identification for word recognition** is more like **thresholded processing** (the lumber mill) or **cascaded processing** (the food web).

### Section 7: The McMurray et al. (2003) Visual World Experiment

- **McMurray, Tanenhaus, Aslin, and Spivey (2003)** published "Probabilistic constraint satisfaction at the lexical/phonetic interface: Evidence for gradient effects of within-category VOT on lexical access" in the *Journal of Psycholinguistic Research*, volume 32, pages 77–97. They tested identification of a **/b/–/p/ continuum** in a **"visual world" display** designed for **eye-tracking**.
- In the **visual world paradigm**, participants look at a screen of pictures while hearing a spoken word, and an eye-tracker records where they look from moment to moment. Eye movements are fast and largely automatic, so they reveal which candidate words a listener is considering while the word is still unfolding.
- Each display contained four pictures. The **target** was a "pear," the **competitor** was a "bear," and the two **filler** items were a "lamp" and a "ship."
    - "Bear" and "pear" form a **minimal pair**, two words that differ in only one sound, here the voicing of the first consonant.
    - The fillers are unrelated to the target, so looks to them give a baseline for random looking.
- The spoken words came from a continuum that varied VOT from 0 to 40 ms in 5 ms steps. Participants used the **mouse** to click the picture most consistent with what they heard, such as the bear or the pear.
- The mouse responses produced a typical, **step-like identification function**. The proportion of /p/ responses is near 0 at 0, 5, and 10 ms, rises only slightly at 15 ms, jumps to about 0.95 at 20 ms, sits at about 0.92 at 25 ms, and stays near 1.0 from 30 to 40 ms.
    - The **category boundary** falls at $17.5 \pm 0.83$ ms of VOT.
    - Judged by the clicks alone, these data match exactly what a threshold model predicts: a sharp jump from one category to the other.

### Section 8: Eye Movements Reveal Gradient Ambiguity

- The eye movements reveal that ambiguity is **gradient**, meaning it changes smoothly with distance from the boundary rather than being all-or-none. Listeners make **more looks to the competitor** when the VOT is **close to the category boundary**.
- To isolate this effect, trials were sorted by which picture the participant finally clicked. Then the proportion of fixations to the competitor picture was tracked over time, from 0 to 2000 ms after word onset.
    - On trials where the participant clicked the **bear**, the VOTs were 0, 5, 10, and 15 ms, and the competitor was the pear. Looks to the pear start at about 0.04, peak

somewhere between 400 and 600 ms, and fall back down by about 1600 to 2000 ms. The 15 ms tokens produce the highest peak, about 0.15, and keep looks to the pear elevated the longest. The 0 ms tokens produce the lowest peak, about 0.11, and drop off fastest.
    - On trials where the participant clicked the **pear**, the VOTs were 20, 25, 30, 35, and 40 ms, and the competitor was the bear. The 20 ms tokens produce the highest peak, about 0.13, and keep looks to the bear elevated the longest. The 40 ms tokens drop off fastest.
- Every trial in one of these groups ended with the same click, so every token in the group was categorized the same way. Even so, looks to the competitor depend on exactly how far the VOT is from the boundary. Differences in VOT **within a category** reach the level of word recognition.
- According to McMurray and colleagues, the effect of VOT falls mainly on the **duration of activation**. Near-boundary tokens do not just raise competitor looks briefly. They keep the competitor active for a longer stretch of time.

**Section 9: Early Versus Late Fixations to the Competitor**

- The effect becomes much clearer when time is split into two windows and fixations are averaged within each one. The **early window** runs from 300 to 1100 ms, and the **late window** runs from 1100 to 1900 ms. Fixation proportion to the competitor is then plotted against VOT, with the category boundary marked at 17.5 ms.
- On trials where the participant clicked the **bear**, which means looks to the pear:
    - In the early window, looks are about 0.075 at 0 ms, about 0.078 at 5 ms, about 0.092 at 10 ms, and about 0.093 at 15 ms.
    - In the late window, looks are about 0.018 at 0 ms, about 0.025 at 5 ms, about 0.038 at 10 ms, and about 0.043 at 15 ms.
- On trials where the participant clicked the **pear**, which means looks to the bear:
    - In the early window, looks are about 0.105 at 20 ms, about 0.085 at 25 and 30 ms, about 0.052 at 35 ms, and about 0.067 at 40 ms, with a small rise at the far end.
    - In the late window, looks are about 0.054 at 20 ms, about 0.025 at 25 ms, about 0.019 at 30 ms, about 0.018 at 35 ms, and about 0.015 at 40 ms.
- Looks to the competitor climb steadily as the VOT gets closer to the boundary from either side, producing a V-shaped pattern that peaks right at the boundary. In the late window, the pattern is especially clean. Looks to the pear at 15 ms are more than twice the looks at 0 ms, and looks to the bear at 20 ms are more than three times the looks at 40 ms.
- Early-window looks are higher overall, since listeners are still considering both pictures at that point. Late-window looks are much lower, since listeners have mostly settled on an answer. The gradient is still present late, which fits the idea that near-boundary tokens keep the competitor active for longer.

**Section 10: Cascaded Activation from Features to Phonemes in the TRACE Architecture**

- The gradient competitor effect fits the idea that activation flows in a **cascaded** way from **features** to **phonemes**. The **TRACE model** of spoken word recognition is built around three layers of processing units.
    - The **feature layer** receives a **pseudo-spectral input representation**, a simplified, spectrogram-like coding of the speech signal, across time slices numbered 0 to 44. It has seven feature dimensions labeled POW, VOC, DIFF, ACU, GRD, VOI, and BUR, where VOI stands for voicing and BUR stands for the burst of air at the release of a stop. Horizontal bars mark which values of each feature are active at each moment, and the cross-hatched region from about 36 to 44 marks the end of the input.

    - The **phoneme layer** contains units for phonemes such as /^/ (the vowel in "but"), /b/, /r/, and /t/. Each phoneme unit is copied many times in overlapping rows, so there is a separate detector for that phoneme starting at each point in time. The boxed units are the most active copies.
    - The **word layer** contains units for whole words, such as "abrupt," written ^ b r ^ p t, and a word written b r i d. Each word is also copied across time, and the bolded, boxed copy of "abrupt" is the one lined up with the input.
    - Arrows run **upward** from the feature layer to the phoneme layer, and both **upward and downward** between the phoneme layer and the word layer.
    - The curved connectors with filled circles beside each layer stand for **lateral inhibition**. Units within the same layer compete, and an active unit suppresses its rivals.
- Even stimuli that are identified as /b/ on 100% of trials can vary in **how strongly and unambiguously** the "voiced" feature is activated. A token with a 0 ms VOT activates voicing very strongly, while a token with a 15 ms VOT activates it more weakly, even though both are labeled /b/.
- If a stimulus is obviously **+bilabial**, meaning it is made with both lips, and obviously **+stop**, meaning airflow is completely blocked and then released, the only open question is voicing. This creates some **momentary ambiguity between /b/ and /p/**, the two bilabial stops.
    - In a cascaded system, the weak voicing activation passes partly to /b/ and partly to /p/.
    - Those partially active phonemes pass activation on to the words "bear" and "pear."
    - That ambiguity shows up in **eye movements** while the participant is deciding where to click.

**Section 11: Partial Activation Reaches the Lexical Level: Andruski et al. (1994)**

- **Andruski et al. (1994)**, published in *Cognition*, volume 52, pages 163–187, found evidence that partial activation can reach all the way to the **lexical level**, the level where words are stored. They found that you can even get **semantic priming** with nonwords.
- **Semantic priming** happens when hearing one word speeds up processing of a word related in meaning. Andruski and colleagues measured it with a **lexical decision task**, where participants decide as quickly as possible whether a target string is a real word. A faster reaction time to the target means the prime had already partly activated it.
- The primes began with voiceless stops. The initial consonant was either **unaltered** or had its VOT shortened by **one third (−1/3)** or **two thirds (−2/3)**, which pushed it toward its voiced counterpart. The unaltered, −1/3, and −2/3 primes were all semantically related to the target, and a set of **unrelated** primes served as the baseline.
    - In **word-to-word** items, the voiced counterpart of the prime is a real word, as when "pet" is pushed toward "bet." For example, "pill" pushed toward "bill" still primes "drug."
    - In **word-to-nonword** items, the voiced counterpart of the prime is a nonword, as when "king" is pushed toward "ging." For example, "beace" primes "war," and "ging" primes "queen."
- Mean reaction times to the targets in Experiment 1 were approximately:
    - **Unaltered** primes: about 858 ms for word-to-word items and about 808 ms for word-to-nonword items.
    - **−1/3** primes: about 857 ms for word-to-word items and about 820 ms for word-to-nonword items.
    - **−2/3** primes: about 888 ms for word-to-word items and about 816 ms for word-to-nonword items.

    - **Unrelated** primes: about 965 ms for word-to-word items and about 945 ms for word-to-nonword items.
- Every related condition is faster than its unrelated baseline, by roughly 75 to 140 ms. In the word-to-nonword items, priming survives even at −2/3, where the target is about 130 ms faster than baseline. In the word-to-word items at −2/3, priming is smaller, about 77 ms, so the advantage shrinks when the altered prime is pushed toward another real word.
- **Lexical representations can be activated even by incomplete phonetic information.** A token of "peace" with a shortened VOT, heard as something like "beace," is not a clean /p/, yet it still activates "peace" strongly enough to speed recognition of "war."

**Section 12: Graded Competitor Looks Are Evidence Against a Threshold Model**

- The graded eye-movement pattern is one piece of evidence for **cascaded processing** in speech perception. In a thresholded model, it would be very hard to explain why the number of looks to the other member of the **minimal pair** increases **gradually** as the VOT approaches the category boundary.
- In the threshold model, the connection between the feature level and the phoneme level is effectively blocked, marked with a red X, until the feature level crosses its threshold. Graded voicing activation never reaches the phonemes, and the phoneme level stays flat until it commits all at once.
    - Under that model, every token categorized as /b/ would produce the same /b/ activation, the same word activation, and the same number of looks to the pear.
    - The data contradict this prediction. In the late window, looks to the pear rise from about 0.018 at 0 ms to about 0.043 at 15 ms, even though every one of those tokens was clicked as "bear."
- In a cascaded model, graded feature activation flows continuously into the phoneme and word levels. A 15 ms token partly activates /p/ and therefore partly activates "pear," which produces more looks to the pear than a 0 ms token does.

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**Section 13: Feedback from the Lexical Layer to the Phoneme Layer**

- The **TRACE model** also includes **feedback** from the **lexical layer** to the **phoneme layer**. The arrows between these two layers run in both directions. Phonemes send activation up to the words that contain them, and active words send activation back down to the phonemes they contain.
    - **Bottom-up** connections carry evidence from the signal upward, from features to phonemes to words.
    - **Top-down** connections carry knowledge of words downward, so an active word strengthens the phonemes that belong to it.
- With feedback, word knowledge can change how a phoneme is perceived. A sound that is acoustically ambiguous between two phonemes can be pushed toward the phoneme that completes a real word. Several experiments provide evidence for this kind of feedback.

**Section 14: Gisses and Kifts for Valentine's Day: The Ganong Effect**

- **Ganong (1980)** studied how **phoneme identification** is affected by **word knowledge**. The question was whether knowing which sound sequences are real words changes where listeners place a phoneme category boundary.
- Ganong built **word/nonword pairs** in which one end of a voicing continuum is a real word and the other end is not, such as **"corpse/gorpse."** "Corpse" begins with the voiceless /k/ and is a word, while "gorpse" begins with the voiced /g/ and is not a word.
    - "Gisses and kifts" is built from the same kind of pairs. In "kiss/giss," the word is at the voiceless end, so a "giss" is the nonword version. In "gift/kift," the word is at the voiced end, so a "kift" is the nonword version.

    - Some continua have the word at the voiced end, labeled **Word–Nonword**, and others have the word at the voiceless end, labeled **Nonword–Word**.
- For each pair, Ganong produced a **continuum** spanning the relevant phoneme boundary in **5 ms steps** of VOT. Listeners reported whether each token began with the voiced or the voiceless consonant.
- Responses are plotted as the **proportion of voiced responses** at six positions along the continuum: the short-VOT endpoint, −7.5 ms, −2.5 ms, +2.5 ms, +7.5 ms, and the long-VOT endpoint, with the boundary falling between −2.5 and +2.5 ms. The **lexical effect** is the difference in voiced responses between Word–Nonword and Nonword–Word continua at each position. A larger lexical effect means word knowledge pulled responses further toward the word.

**Section 15: Prediction One: A Late "Correction" Process**

- One possibility is a late **"correction" process**. Categorical perception works as usual, but afterward there is a **bias to report having heard a word**. The bias operates on the response, not on perception itself.
- Under this model, the **lexical effect is flat** at about 0.25 across the whole continuum. The bias adds the same amount everywhere, no matter how clear or ambiguous the sound is.
    - The **Word–Nonword** continuum gives voiced responses of about 1.00 at the short-VOT end, 0.97 at −7.5 ms, 0.90 at −2.5 ms, 0.45 at +2.5 ms, 0.28 at +7.5 ms, and 0.25 at the long-VOT end.
    - The **neutral** baseline, where neither end has a lexical advantage, gives about 1.00, 0.96, 0.73, 0.27, 0.04, and 0.00.
    - The **Nonword–Word** continuum gives about 0.75, 0.72, 0.65, 0.20, 0.03, and 0.00.
- Subtracting the Nonword–Word value from the Word–Nonword value gives the lexical effect. At the short-VOT end it is 1.00 minus 0.75, which is 0.25. At −2.5 ms it is 0.90 minus 0.65, which is 0.25. At the long-VOT end it is 0.25 minus 0.00, which is 0.25.
- A flat lexical effect means that even a perfectly clear endpoint token gets reported as the word on an extra quarter of trials. A clear "gorpse" would sometimes be reported as "corpse" simply out of a preference for words.

**Section 16: Prediction Two: Word Knowledge Shifts the Category Boundary**

- A second possibility is that word knowledge works to **shift the categorical boundary**. Under this model, the word changes where the boundary falls rather than adding a fixed bias to every response.
- Under this model, the lexical effect is near 0 at both endpoints, about 0.1 at −7.5 and +7.5 ms, and peaks at about 0.38 at −2.5 and +2.5 ms.
    - The **Word–Nonword** continuum gives voiced responses of about 1.00, 0.99, 0.88, 0.50, 0.12, and 0.01.
    - The **neutral** baseline is the same as before: about 1.00, 0.96, 0.73, 0.27, 0.04, and 0.00.
    - The **Nonword–Word** continuum gives about 0.99, 0.88, 0.50, 0.12, 0.01, and 0.00.
- The two lexical curves are copies of the neutral curve slid sideways in opposite directions. At −2.5 ms the lexical effect is 0.88 minus 0.50, which is 0.38. At −7.5 ms it is 0.99 minus 0.88, which is 0.11. At the endpoints it is essentially 0.
- The lexical effect is **strongest for the items closest to the category boundary**. Those are the stimuli for which lexical information can help **resolve ambiguity about the initial phoneme**. A clear endpoint token needs no help from word knowledge, so the word has almost no effect there.

**Section 17: Ganong's Results**

- The results are strikingly consistent with the **second model**, in which word knowledge shifts the boundary.
    - The **Word–Nonword** continuum gives voiced responses of about 0.98 at the short-VOT end, 0.95 at −7.5 ms, 0.67 at −2.5 ms, 0.28 at +2.5 ms, 0.14 at +7.5 ms, and 0.11 at the long-VOT end.
    - The **Nonword–Word** continuum gives about 0.96, 0.83, 0.34, 0.09, 0.02, and 0.02.
- The lexical effect is about 0.02 at the short-VOT end, rises to about 0.12 at −7.5 ms, peaks at about 0.33 at −2.5 ms, falls to about 0.19 at +2.5 ms and about 0.12 at +7.5 ms, and ends at about 0.09 at the long-VOT end.
    - At the peak, 0.67 minus 0.34 gives the lexical effect of 0.33.
    - The effect is small at the endpoints and largest next to the boundary, which is the signature of a boundary shift rather than a constant bias.
- Ganong raised a number of possibilities for how this could happen, focusing on different kinds of **guessing strategies**. Listeners might guess the word when they are unsure about an ambiguous sound.
- Later models emphasize the potential role of **feedback** in these shifts. Activation from the word level flows back down and strengthens the phoneme that completes the word, which moves the boundary at the level of perception itself.

**Section 18: A Simulation of the Ganong Effect in the TRACE Model**

- TRACE simulates the Ganong effect with an input whose first segment is **ambiguous between /b/ and /p/**, followed by /l/, /^/, and /g/. The only real word this input can match is **"plug,"** while the /b/ reading leads only to nonwords. Activation at the phoneme level and the word level is tracked at four points as the input unfolds, with silence marked by a dash at each end of the input.
- **Frame 1**, at the ambiguous first segment:
    - At the phoneme level, the unit for the preceding silence is still high at about 0.6, /b/ and /p/ are nearly tied at about 0.2, and a weak /d/ unit sits near 0.1.
    - At the word level, only the silence unit is active, at about 0.17.
- **Frame 2**, as /l/ arrives:
    - At the phoneme level, /b/ and /p/ are still nearly tied at about 0.42, /l/ is at about 0.45, /r/ is at about 0.37, and /^/ is starting to rise at about 0.25.
    - At the word level, the silence unit is at about 0.33, and a large cluster of candidate words starting with /b/ or /p/ sits between about 0.05 and 0.12.
- **Frame 3**, as /^/ arrives:
    - At the phoneme level, /l/ and /^/ are at about 0.5, /g/ is at about 0.45, /p/ is just above /b/ at about 0.43, /r/ is at about 0.37, /k/ is at about 0.25, /a/ is at about 0.17, and /d/ is at about 0.08.
    - At the word level, **"plug"** has risen to about 0.28. Words such as **"blood"** and **"blush"** are also active at about 0.12 to 0.17, since they match the beginning "bl-" plus the vowel.
- **Frame 4**, as the final /g/ arrives:
    - At the phoneme level, /^/ is at about 0.63, /g/ is at about 0.6, /l/ is at about 0.57, and **/p/ has pulled ahead of /b/**, about 0.47 versus about 0.40.
    - At the word level, **"plug"** is above 0.45 and is now the most active word, while the other candidates have dropped to about 0.07 to 0.1.
- As the input unfolds over time, activity **accumulates for "plug,"** especially when the final /g/ arrives. The /g/ is highly inconsistent with "blood" and "blush," so those competitors fall away and "plug" wins.
- TRACE explains word effects on phoneme category boundaries in terms of **top-down processing**. Activity at the lexical level produces **feedback** to the phoneme level, which

produces **greater activation for the phoneme consistent with a word**.
    - The acoustic evidence for /b/ and /p/ is the same throughout.
    - The growing activation of "plug" sends activation back down to /p/.
    - That added activation pushes /p/ above /b/, so the ambiguous sound is perceived as /p/.

**Section 19: The Interactive Activation Model**

- The **TRACE model** is a version of the **Interactive Activation model** for **written word recognition**, developed by **McClelland and Rumelhart (1981)**.
- The Interactive Activation network has three levels of units.
    - At the **feature level**, units detect simple line segments, such as horizontal, vertical, and diagonal strokes.
    - At the **letter level**, units stand for letters in a given position, such as D, K, and R in the fourth position of a word.
    - At the **word level**, units stand for whole words, such as WEAR, WORK, WORD, and WEAK.
    - Excitatory connections link consistent units across levels, so a K unit supports WORK and WEAK. Inhibitory connections link rival units within a level, so word units compete with each other and letter units compete with each other.
- A hypothetical set of features extracted on one trial of a word perception experiment consists of the letters W, O, and R, followed by a fourth letter that is partly covered by ink blots. The visible strokes of the fourth letter fit either an R or a K.
    - The letter level alone cannot decide between R and K.
    - The word level settles the question: WORK is a word and WORR is not, so the WORK unit becomes active and sends activation back down to K.
    - The fourth letter is then perceived as K, which is the written version of the same top-down effect seen in the Ganong effect.
- Many of the principles are the same in both models, including excitation between levels, competition within levels, and feedback from words to smaller units. A key issue in speech perception is the way the **stimuli unfold over time**. A written word is available all at once, while a spoken word arrives one sound after another. TRACE handles this by copying every feature, phoneme, and word unit across time positions.

**Section 20: Categorical Perception and the "Lexical-Phonetic Interface"**

- Perception appears to be shaped by **category boundaries**. Given the same physical difference, it is more difficult to distinguish two sounds from the **same category** than two sounds from **different categories**.
- The findings on cascaded processing and feedback do not undermine that basic finding. They add two further facts about how phonemes and words relate.
    - **Ambiguity about phoneme identity affects word recognition.** Near-boundary VOT values produce more looks to the competitor word, and altered primes still activate the meanings of words.
    - **Word-level information affects phoneme categorization.** In the Ganong effect, the boundary shifts toward whichever end of the continuum forms a real word.
- The word recognition system therefore cannot just sit and wait for the phoneme recognition system to finish its job, the way a furniture factory waits for lumber from the mill. **Spoken word recognition is more interactive** than that. Information flows continuously between levels, both upward and downward.

**Section 21: Spoken Word Recognition Is Fast: Allopenna et al. (1994)**

- **Allopenna et al. (1994)**, published in the *Journal of Memory and Language*, used eye-tracking to measure how quickly spoken words are recognized. **Spoken word recognition is also very fast.**
- Participants saw a 5-by-5 grid with a fixation cross in the center. Four pictures surrounded the cross: a **beaker**, a **beetle**, a **speaker**, and a baby **carriage**. Four geometric shapes, a triangle, a square, a circle, and a diamond, filled other cells.
    - The **referent** is the word actually spoken, "beaker."
    - The **cohort** competitor, "beetle," shares the **beginning** of the word, the sounds "bea-."
    - The **rhyme** competitor, "speaker," shares the **end** of the word, "-eaker."
    - The **unrelated** item, "carriage," shares no sounds with the referent.
- The probability of fixating each picture type was tracked from 0 to 1000 ms after the spoken word began, in the full competitor condition of Experiment 1. The data are averaged over all stimulus sets, and beaker, beetle, speaker, and carriage are the examples from one set.
    - **Referent** looks start rising at about 200 ms, reach about 0.2 at about 400 ms, about 0.5 at about 600 ms, and level off near 0.95 from about 850 ms onward.
    - **Cohort** looks also start rising at about 200 ms and climb right alongside the referent at first. They peak at about 0.18 around 500 to 550 ms and then fall to near 0 by about 800 ms.
    - **Rhyme** looks start rising a bit later, at about 250 to 300 ms, peak lower at about 0.1 around 450 to 700 ms, and fall away by about 850 ms.
    - **Unrelated** looks stay near 0 the whole time, with only a slight rise to about 0.03 around 600 to 750 ms.
- The **average target offset**, the point where the spoken word ends, falls just under 400 ms. Referent looks are already climbing before the word has finished, so recognition begins while the word is still arriving.
    - Early on, "beaker" and "beetle" match the input equally well, so the referent and cohort rise together.
    - Once the later sounds rule out "beetle," cohort looks drop and referent looks keep climbing.
    - The rhyme is also activated, though less strongly, even though it mismatches at the very first sound, so partial matches anywhere in the word attract some activation.

**Section 22: TRACE Simulations of the Eye-Movement Data**

- TRACE's predicted activations can be converted into **predicted fixation probabilities** and placed alongside the actual fixation data from 0 to 1000 ms.
    - For the **referent**, TRACE rises slightly earlier than the real fixations. At about 600 ms, TRACE predicts about 0.6, while actual fixations are about 0.5. Both curves converge near 0.95 by about 800 ms.
    - For the **cohort**, TRACE peaks at about 0.17 around 480 ms, and actual fixations peak at about 0.18 around 500 to 550 ms. Actual cohort looks stay elevated a little longer, at about 0.14 until about 650 ms, before both curves fall to near 0 by about 800 ms.
    - For the **rhyme**, TRACE and the actual fixations both peak at about 0.1 between about 450 and 600 ms, and both decline slowly to about 0.03 by 1000 ms.
- The simulation matches the data at a fine **granularity**, tracking the rise and fall of each competitor over tens of milliseconds. Eye movements provide a moment-by-moment estimate of the changes in **"activity"** for different competitors over time. They make **cognitive processing visible before a final response is issued**.

- The graded effects of **category goodness** in the categorical perception task follow the same logic. A final label or click hides differences in how good an example of a category a sound is, while moment-by-moment measures such as eye movements reveal those graded differences in activation.